



PROFESSIONAL PRACTICES

ASSIGNMENT 3

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Assignment: _____

Date: _____

Question # (01)

a Advantages of Not requiring signature:

(i)

Faster Transaction:-

Skipping signatures speeds up checkout especially in busy places like gas stations and supermarkets.

(ii)

Lower Operational Cost:-

Retailers save time and money by reducing paperwork and staff involvement.

(iii)

Technology based Security:

Modern system use PINS, EMV chips, and transaction monitoring which are often more secure than signatures.



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Arguments against not requiring a signature:-

(i)

Higher Fraud Risk:

Without signature, it becomes easier for someone to misuse a stolen card.

(ii)

Lack of Verification:

Signature can act as an extra identity check, especially for high value purchases.

(iii)

Dispute difficulty:

It may be harder to resolve disputes if no signature record exists

Opinion:-

Retailers should not always require a signature, especially for small value or low risk-transaction. Modern digital security methods are often more reliable.



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Question # 02

Solution:-

Requiring both a fingerprint and a password is not redundant. It is actually a strong security practice known as multifactor authentication.

Reasons:

(i)

Layered Security:

Fingerprint: Something you are

Password: Something you know.

Using both reduces the chance of both unauthorized access.

(ii)

Fingerprint Limitation:-

Fingerprint can be copied in rare cases, and they cannot be changed if compromised.

(iii)

Backup Protection:-



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If the fingerprint reader fails or is damaged, the password still allows access.

(iv)

Higher Security of sensitive data:-

ISPs handle personal and financial data, so extra protection is justified.

Conclusion:-

Requiring both fingerprint and password is a good security decision, not pointless. It significantly improves protection against hacking, identity theft and unauthorized access.

ASSIGNMENT # 3

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-: QUESTION No. 3 :-

:Reducing Online Banking Scams with a ".bank" Domain:

Online banking scams have become increasingly common due to phishing, pharming, and (~~unidentified~~) identity theft. Fraudsters often create fake websites or (~~domain~~) manipulate DNS to trick users into revealing sensitive information. One proposed solution is a specialized Internet domain, ".bank", restricted exclusively to verified chartered banks. This aims to provide a secure and trustworthy way for users to access banking websites.

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Techniques Prevented by the ".bank" Domain:-

•Pharming:

- Pharming ~~is~~ involves redirecting users from legitimate websites to fraudulent ones by manipulating DNS or using fake URLs.

- By limiting the ".bank" domain to verified banks, attackers cannot register fake domains, reducing the risk of pharming attacks.

•Website-based Phishing:

- Phishing scams trick users into providing login credentials through fake websites

linked in emails and messages.

- Users can check that a website ends with ".bank" to verify authenticity, making phishing scams less effective.

Techniques Not Prevented by the ".bank" Domain:-

• Smishing & Vishing:

- These attacks occur via SMS (smishing) or voice calls (vishing) and do not rely on websites URLs.
- A restricted domain cannot prevent these social engineering attacks.

• Insider Fraud or Malware Attacks:

- Attacks by malicious insiders or malware on a user's device bypass website security entirely.
- The ".bank" domain does not protect against these internal or device-level threats.

Evaluation:-

• Advantages:

- Enhance user trust by providing a verified namespace for banks.
- Reduces phishing and pharming risks.

• Limitation:

- Cannot prevent SMS/voice-based frauds or insider attacks.
- Needs to be combined with other security measures such as multi-factor authentication, user education, and system monitoring.

Conclusion :-

This ".bank" domain is a strong step towards reducing online banking scams. While it is not a complete solution, it significantly improves security for web-based banking interactions and, when combined with other protective measures, can reduce fraud and protect sensitive user data.

-: QUESTION No. 4 :-

: Click Fraud and Digital Forgery :

The digital age has introduced new forms of cybercrime. Click Fraud and Digital Forgery are two prominent examples, exploiting digital systems for financial or legal gain. Understanding these threats is essential for IT professionals to prevent misuse.

⇒ Click Fraud:-

• Definition:

Click fraud is the deliberate act of repeatedly clicking on online advertisements to generate false revenue or to deplete a competitor's advertising budget.

• Example:

A competitor uses automated bots to click thousands of times on a company's pay-per-click (PPC) ads. The company is charged for each despite no real customer engagement.

• Impact:

- Wastes advertising budgets.
- Misleads business about ad effectiveness.
- Undermines trust in online advertising systems.

⇒ Digital Forgery:-

• Definition:

Digital forgery involves creating or altering digital content to deceive others. This can include modifying electronic documents, images, or contracts.

• Example:

A fraudster edits a PDF contract to change

payment amounts or add unauthorized clauses, presenting it as an official document.

• Impact :

- Causes financial losses.
- Leads to legal disputes.
- Erodes trust in digital records and communication.

• Comparison :-

Aspect	Click Fraud	Digital Forgery
• Target	Online advertising systems.	Digital content / documents
• Method	Automated or repeated clicks.	Editing/modifying digital files.
• Purpose	Generate false revenue / harm competitor.	Deceive others for financial / legal gain.
• Example	Bots clicking PPC ads.	Modifying a PDF contract.

• Conclusion :-

- Click fraud exploits online advertising systems, while digital forgery exploits trust in digital content.
- Both are cybercrimes but operate in different domain: click fraud affects revenue, and digital forgery affects legal and financial trust.
- Awareness of these techniques is crucial for IT professionals to protect organizations and users from digital exploitation.

Question No 5

Compare Competitive Intelligence and Industrial Espionage

Competitive Intelligence and Industrial Espionage both deal with collecting information about competitors, but they are completely different in terms of legality, ethics and methods.

Competitive Intelligence is a process of collecting and analyzing information about competitors in a legal and ethical manner. The information is gathered from publicly available sources such as company websites, annual reports, market research, social media advertisements, customer feedback and the news articles. The main purpose of competitive intelligence is to help organization understand the market trends, competitors strengths and customers needs so that they can make better decision. Competitive Intelligence is widely accepted and encouraged in professional business practices.

On the other hand, **Industrial Espionage** refers to practice of obtaining confidential or secret information through illegal and unethical means. They may include hacking

computer system stealing trade secrets, spying, bribing employees or using surveillance devices. The goal of industrial espionage is to gain an unfair competitive advantage. Such activities can cause serious damage to organizations and may result into legal penalties, heavy fine and loss of reputation.

Conclusion: In conclusion competitive intelligence is a legitimate and responsible business activity, while industrial espionage is a criminal act that violates laws and ethical standards. Organizations should always rely on competitive intelligence rather than illegal methods.

Question No 6:-

Computer/system
technical Failure and suggestions
for improvement:

on assignments the system becomes slow unresponsive and sometimes restarted automatically without warning this resulted in loss of unsaved data increased stress and wasted time.

after analyzing the problem, I found that the main causes were overheating, limited RAM, running too many background applications and outdated system software. poor system performance directly affected productivity and academic work.

To make the system better several improvements can be suggested. first Software applications should include the strong auto-save feature so that user data is protected even during unexpected failures. Second, the operating system should have better memory and task management to prevent freezing. third hardware design should notify users before automatic updates or reboots.

Lastly regular system maintenance such as updates, antivirus scanning and removal of unnecessary programs should be encouraged.

in conclusion technical features can significantly effect work efficiency, with improved system design better soft were features and proper user practices such features can be minimized leading to a more reliable and user-friendly computing system...